D2.1 Cell and nuclear division: data-based questions

NAME | DATE | TOTAL _____ / 9

Read the data carefully before you answer. The number of marks shows how many separate points to make. Show your working in calculations.

1 A drug that targets mitosis

9 MARKS

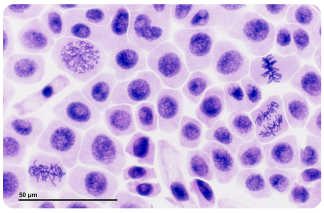
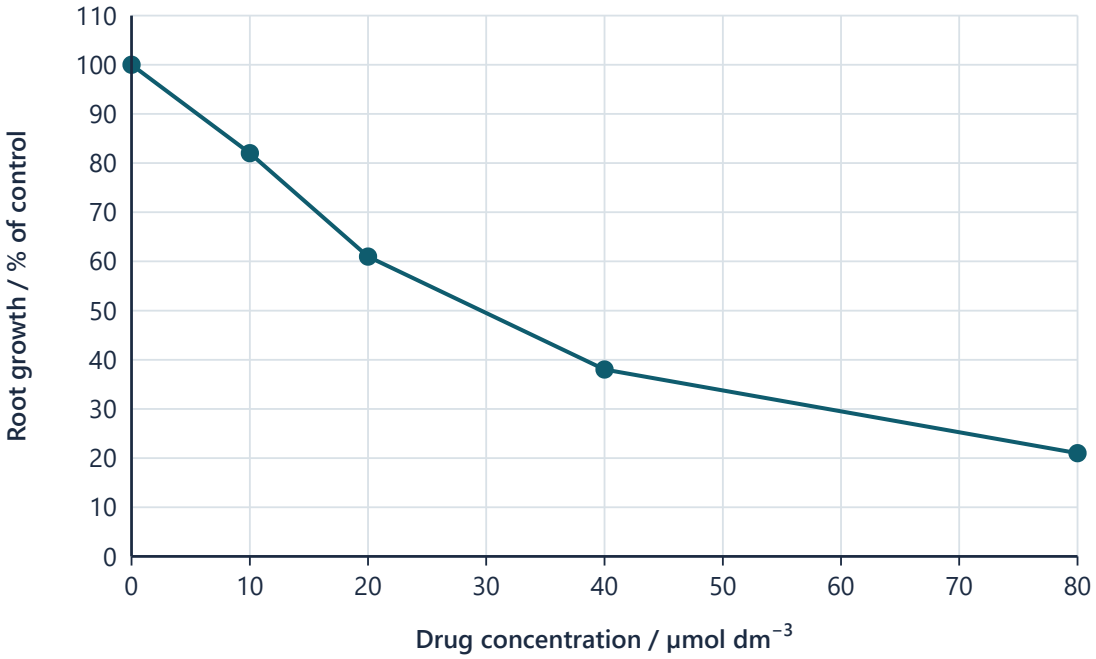


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Onion root tips were grown for 24 hours in different concentrations of a drug being tested as an anticancer treatment. Root growth was measured, and cells from the tips were stained and counted.

DRUG / MMOL DM^{-3}	0	10	20	40	80
MITOTIC INDEX / %	12	15	19	24	27
CELLS IN ANAPHASE PER 100 CELLS IN MITOSIS	21	14	8	4	2



a STATE the root growth, as a percentage of the control, at 40 μmol dm^{-3}. [1]

b At 80 μmol dm^{-3}, 54 of 200 cells in one field of view were in mitosis. **Calculate** the mitotic index. [1]

c DESCRIBE the relationship between drug concentration and mitotic index. [1]

d EVALUATE the hypothesis that the drug stops cells from completing mitosis. [3]

e SUGGEST how the drug could prevent cells from entering anaphase. [1]

f STATE one advantage and one disadvantage of using plant root tips to test drugs meant for human cancer. [2]